

```
def RobustCMA(D, budget, bounds):  
     $\lambda = 4 + 4 * D$   
    stagnation_threshold =  $30 * \lambda / D$   
  
    # Фаза разведки (20%)  
    es1 = IPOPCMA(D,  $\lambda$ , budget*0.2)  
    m1,  $\sigma$ 1 = es1.run()  
  
    # Детекция застоя  
    if es1.stagnation > stagnation_threshold:  
        # Hybrid Restart  
        top5 = LBFGS(m1, budget*0.1)  
        immigrants = uniform(bounds,  $\lambda // 5$ )  
        m2 = elite_merge(top5, immigrants)  
    else:  
        m2 = m1  
  
    # Фаза эксплуатации (70%)  
    es2 = ActiveCMA(D,  $\lambda$ , budget*0.7,  
m_start=m2)  
    return es2.run()
```